

# Understanding mathematics through resolution of paradoxes

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*“Everything I tell you is a lie”*

## 1. Development of mathematics through paradoxes.

Brain-challenging puzzles attract people for a very long time. For instance, one finds them in the Egyptian Rhind Papyrus written around 1650 B.C. Paradoxes constitute a special type of puzzles aiming to emphasize certain inconsistency, a sort of circularity, and self-contradiction. The earliest known mathematical paradox is the *Cretan Paradox*, dating back to the 6th century B.C. attributed to the semi-mythical Cretan poet Epimenides.

*Epimenides the Cretan says “All Cretans are liars.”*

This statement, because it was uttered by a Cretan, is true if and only if it is false. This oldest paradox is also known to be especially troubling. Its 20th century restatement, the *Barber Paradox*, was given by English philosopher and logician Bertrand Russell in 1903.

*In a town, the barber shaves all and only those who do not shave themselves. Does the barber shave himself?*

Main feature of these logical paradoxes is that they are self-referential. “Drawing Hands,” composed by M. C. Escher in 1948, provides a visual analogy for this type of paradoxes.

The Russell paradox appeared at time of axiomatization of the set theory, in connection with Georg Cantor’s belief that virtually any description of objects will define a set. Russell and Zermelo showed that defining sets which contain themselves as elements would lead to contradictions. These paradoxes have influenced the development of propositional and fuzzy logic.

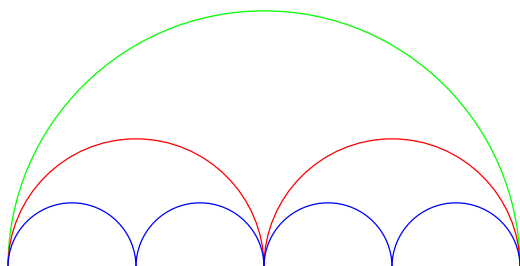
Of the about 40 paradoxes attributed to Zeno (5th century B.C.) the most famous are on the subject of motion.

*There is no motion, because that which is moved must arrive at the middle before it arrives at the end, and so on ad infinitum.*

In other words, a runner must first run one-half, then an additional one-fourth, then an additional one-eighth, etc, always remaining short of his goal. Hence, Zeno concludes, he can never reach it! And here is Aristotle’s criticism: “Zeno’s argument makes a false assumption when it asserts that it is impossible to traverse an infinite number of positions in a finite time”. A resolution of Zeno’s paradoxes undoubtedly contributed to the development of such concepts as the sum of an infinite series, the limit, and establishing the calculus. The idea of limit however was successfully used already by Archimedes who calculated the area of a circle by means of a sequence of inscribed regular polygons.

At the same time, there is a number of paradoxical limiting processes, as the following one

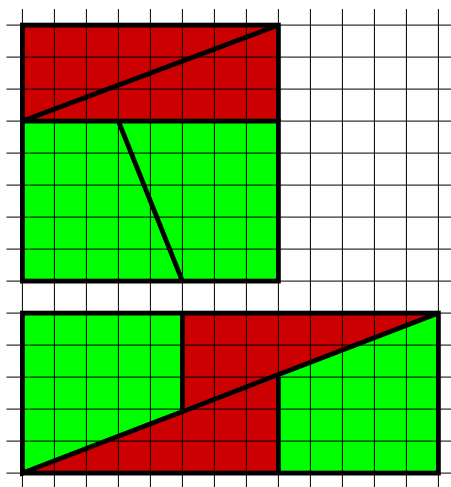
*Let the biggest semicircle have the radius 1, thus the diameter 2. The biggest semi-circumference is  $\pi$ . The next two smaller semi-circumferences add up to  $\pi$  again. The next four even smaller semi-circumferences still add up to  $\pi$ . One can show that at each iteration the circles become smaller but the sum remains  $\pi$*



*In the limit each of the the circles contracts to a point, and one obtains the segment of length 2. This proves  $\pi = 2$ .*

Mental games with *infinity* are proven to be *dangerous* and often lead to surprises. Fractals are known as objects resulting from an infinite self-similar reproduction have some *strange* properties. For instance, Koch snowflake has finite area and infinite perimeter at the same time. Another fractal called Sierpinsky carpet has zero area and yet has a non-empty interior.

Some paradoxes rest only on visual illusions. For instance, the next rearrangement of  $8 \times 8$  square into  $5 \times 13$  rectangle *proves* that  $64 = 65$ .



But the sizes of the square and rectangle are not random! Note that this particular visual illusion is based on the property of Fibonacci numbers (1,1,2,3,5,8,13,...)

$$|F_k^2 - F_{k-1}F_{k+1}| = 1.$$

This implies smallness of the difference of the slopes  $|F_{k-1}/F_k - F_k/F_{k+1}|$ .

In that geometrical paradox the contradiction is only apparent. The following *Missing Dollar Paradox* may even be considered as an arithmetic joke, because so simple solution it has.

*Three traveling salesmen break down and are forced to spend the night at a small town inn. They go in and the innkeeper tells them, "The rooms are \$30". Each man pays \$10 and they go up to the room. The husband of the innkeeper says to her, "Did you charge them the full amount? Why not give them five bucks back since their car is broken and they hadn't planned to stay here." She then brings the men five \$1 bills and each man takes one while the other \$2 rest on the table. Originally each man paid ten dollars ( $10 \times 3 = 30$ ), now each man has paid nine dollars ( $9 \times 3 = 27$ ) there are \$2 sitting on the counter ( $27 + 2 = 29$ ). The last dollar had disappeared.*

Solution: amount that salesmen spent is  $(10 - 1) \times 3 = 27$ ; amount that innkeeper received is  $(30 - 5) + 2 = 27$ . There is no missing dollar.

The next one, known as an *Unexpected Exam Paradox* is however more elegant.

*A professor announces that he is giving an examination some day the week after, and the exact day of the examination will surprise his students. One student reasons about the professor's statement as follows: Suppose the exam is on Friday (the last day of the week), then on Thursday night, the students will know that the exam will be on the next day, so there is no surprise. Hence the exam will NOT be on Friday. Now suppose the exam is on Thursday, then on Wednesday night, they will all know that the exam is on the next day: again there is no surprise and therefore the exam will not be on Thursday. The same argument applies to Wednesday, Tuesday and finally Monday, leaving the conclusion that there will not be any exam in the following week.*

*The sequel to this story is that when this student receives the exam paper, he is very surprised.*

The *Uninteresting Number Paradox* is our next brainteaser.

*Is it possible to partition the set of whole numbers into two subsets: interesting and non-interesting? If you start to do that, the first number you claim to be uninteresting will be interesting as the first uninteresting number!*

In a way this paradox illustrates, as some philosophers argue, the impossibility of complete axiomatization of mathematics. We mean here David Hilbert's idea to create a perfect artificial language for reasoning, for doing mathematics, for deduction. Hence, he stressed the importance of the axiomatic method, whereby one works from a set of basic postulates (axioms) and well-defined rules of deduction to derive valid theorems. The notion of doing mathematics that way goes back to the ancient Greeks and particularly to Euclid and his geometry, which is a beautifully clear mathematical system.

But then in 1931 Kurt Gödel showed that it couldn't be done, that no formal axiomatic theory could contain all of mathematical truth, that they were all incomplete. And this exploded the normal Platonic view of what math is all about. How could one use mathematical reasoning to show that mathematical reasoning is in trouble and has limitations?!

We may continue further our list of paradoxes, but all we aimed to show is their profound impact on the development of mathematics and human thinking, which hardly was doubted.

What needs to be addressed now is the significance and possibilities to incorporate paradoxes in teaching of mathematics.

## 2. Teaching mathematics through paradoxes.

Does the study of paradoxes help student's understanding of mathematics?

The answer to this question is not obvious. I wish to say yes because of a number of reasons.

### Reason 1

The word *paradox* comes from the Greek, meaning *beyond belief*. In cognitive psychology a system of beliefs is called *schemata*. Thus *paradox* is an item of information that contradicts your current schemata, creating a cognitive conflict and disbalance. *Equilibration* is a cognitive process by which the learner attempts to understand an item of information, to restore the balance by either

- interpreting the new experience in terms of the existing schemata (assimilation),

or

- modifying the existing schemata (accommodation).

This way a creation of knowledge for yourself – an understanding – occurs. This restructuring of knowledge makes resolution of paradoxes similar to the solution of Insight problems.

### Reason 2

Resolution of paradoxes is always an effort to overcome one's *mental set* or *functional fixedness*, the state when problem solvers rely too heavily on their previous concepts, expectations, strategies and approaches. One needs a fresh look, a new concept, a different method to show that either

- the contradiction is only apparent,

or

- the paradox rests on invalid or unreasonable grounds.

### Reason 3

Paradoxes give us a warning against formal use of mathematical concepts and formal generalizations. They suggest that meaning has priority over form.

On the other hand, not less serious arguments can be brought against the utility of paradoxes in teaching.

Paradoxes are sort of obstacles, and thus may cause all possible negative reactions such as aggression (which can be turned in good if converted into problem solving mode), avoidance, wrong set or fixedness, even neurotic reaction resulting from self-contradiction, which is the nature of a paradox.

Some philosophers had argued, referring to certain paradoxes, that the world by itself is contradictory — and no solution could therefore be given to those paradoxes.

It is crucial whether one believes that the nature is paradoxical or that any paradox is such only due to restrictive frameworks in which one is trying to make sense of it. The later point assumes that the paradox can and should be resolved by embedding it in a wider context and possibly rethinking the meaning of the situation.

My experience of talking about paradoxes with university students has significant variations. There are students who definitely and undoubtedly learn a great deal from thinking about contradictions, students who this way invent and rediscover for themselves important mathematical constructions and notions. Students who understand the essence of a

puzzling statement, who become curious about the true reasons for it and appreciative for its various interpretations and explorations. How many of such students I have had? To tell the truth, only very few names come to mind.

There is another group of students, for whom the craft of learning from paradoxes is not fully accessible. They still can be good students, hard working, good solvers of standard tests, interested in mathematics, enjoying simple puzzles, but surprisingly helpless with problems like the Missing Dollar Paradox or “ $64=65$ ” paradox (see above). Is it a lack of critical thinking? Not understanding when formal proof becomes a fallacy? Inability to decompose a long argument into more elementary statements?

The large group of students evidently don’t benefit from being directly exposed to paradoxes, they are getting misled, confused, frustrated. An attempt to show how one can prove that *all horses are of the same color* by the method of mathematical induction (if used incorrectly!) makes some students to admit that “I never fully trusted this method!” The instructor’s intention is, using the Horses Paradox, to force the students to find the gap in the “proof”, instead some students are left with the feeling that *the method sometimes works but sometimes does not*. The majority of students have developed (over many years in high school) a habit of “safe learning”, mostly in a declarative mode, using explicit rules, given formulae, memorized facts. Even at this level for some learners there is enough confusions and misunderstandings. There is enough paradoxes resulting from their own misconceptions and psychological obstacles to accept certain mathematical statements. At this point there is no room for critical learning of new material. Therefore such a learner is reluctant to be exposed to further contradictions. He/she does not have a way or habit to resolve them. For such a student often an individual assistance is required before he/she starts to accept the very idea of creative thinking.

These observations suggest that an instructor must be very careful and sensitive while implementing paradoxes into a teaching process. It is a big challenge and always a risk, but it can be a valuable exercise in the problem solving orientations, if used deliberately and cleverly.

### 3. Conclusion.

In this article I attempted to show that mathematical paradoxes carry a special challenge, a hidden message, thinking of which provides the learner with an opportunity to refine their incomplete understanding.

Therefore, of great interest to cognitive sciences would be a more detailed investigation of the influence of instructor’s extensive and creative use of mathematical paradoxes on students’ attitudes and performance. Such research is not only of academic interest but is called by pedagogical practices.